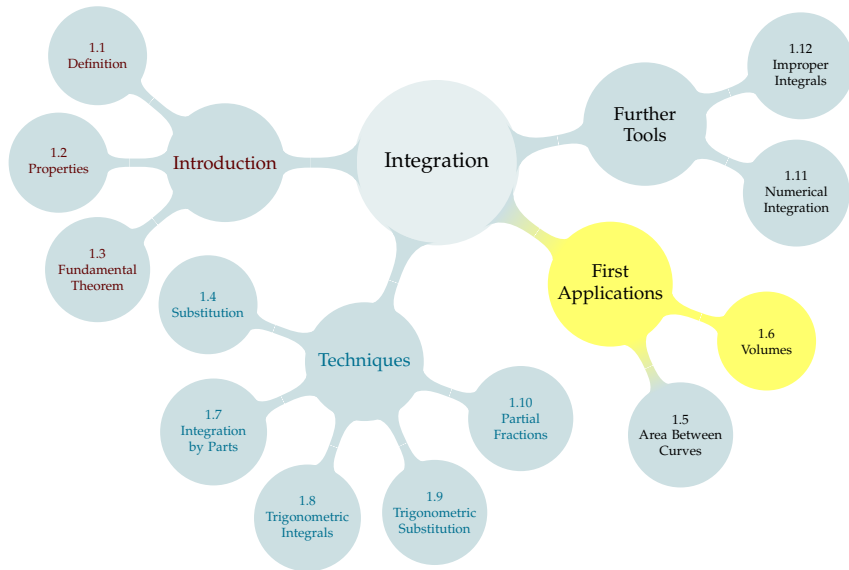
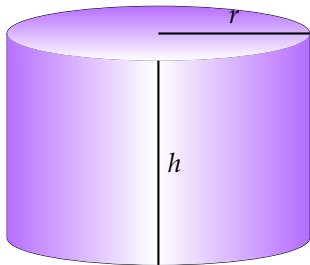


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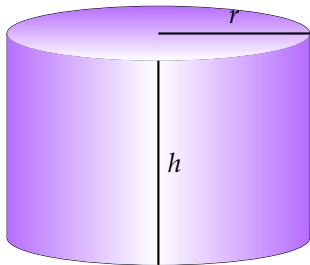


QUICK REFRESHER: VOLUMES OF CYLINDERS

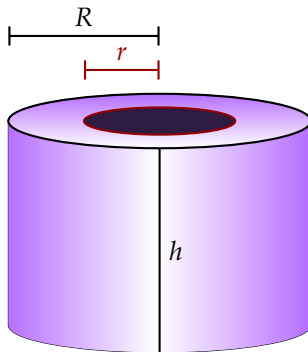


The volume of a cylinder with radius r and height h is:

QUICK REFRESHER: VOLUMES OF CYLINDERS



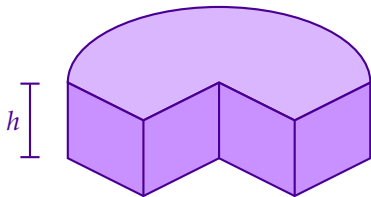
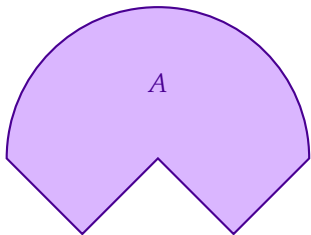
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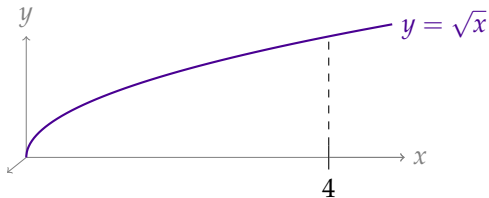
The volume of a washer, with outer radius R , inner radius r , and height h is:

QUICK REFRESHER: VOLUMES OF CYLINDERS

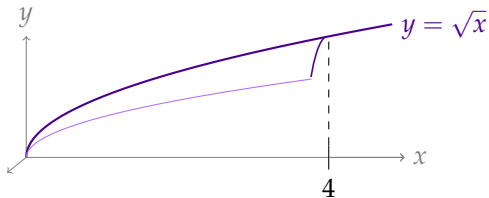
More generally, if we have a shape of area A , and we extrude it into a solid of height h , the resulting solid has volume:



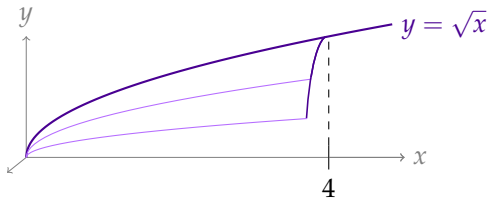
Consider the volume, V , enclosed by rotating the curve $y = \sqrt{x}$, from $x = 0$ to $x = 4$, around the x -axis.



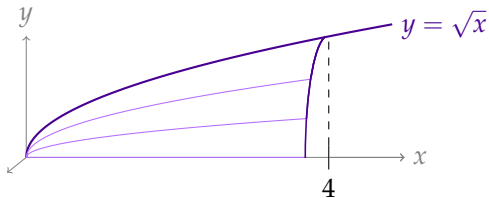
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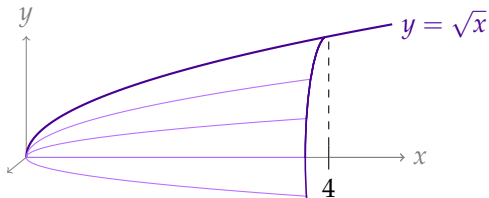
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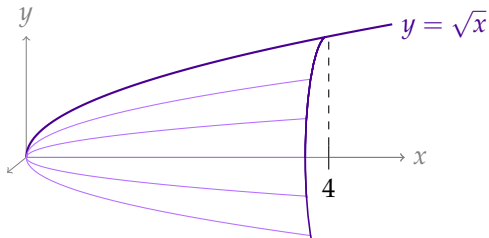
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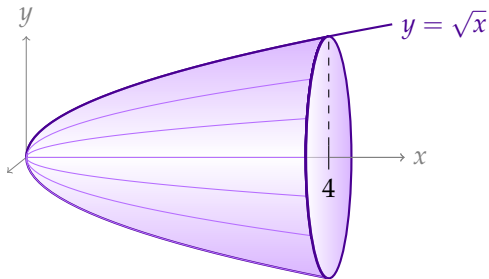
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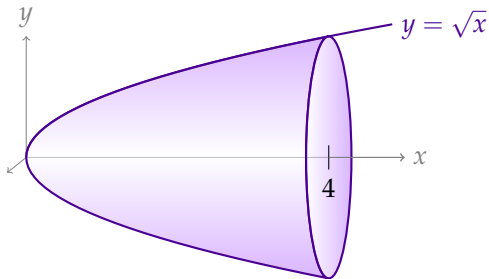
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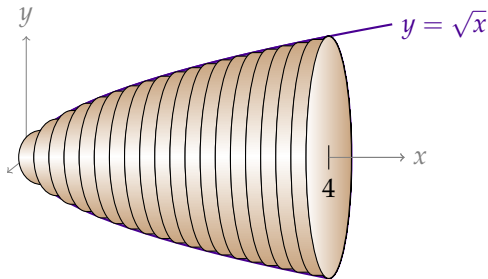
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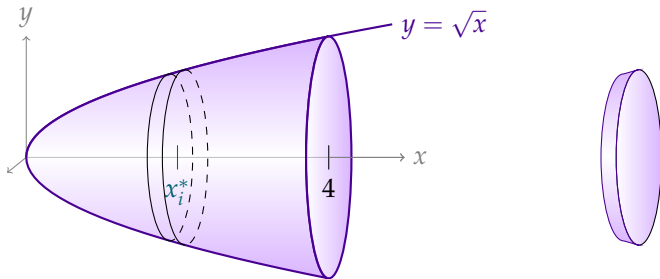


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We cut the solid into slices, and approximate the volume of each slice.

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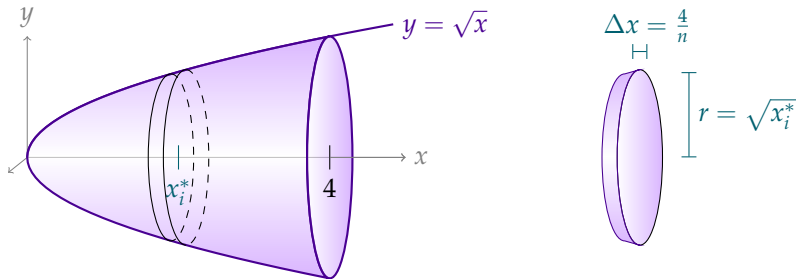


We cut the solid into slices, and approximate the volume of each slice. Each thin slice is *approximately* a cylinder.

If we use n slices, the width of each is:

The radius of the slice at $x = x_i^*$ is:

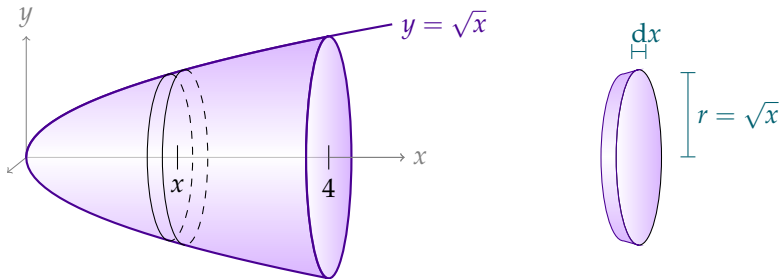
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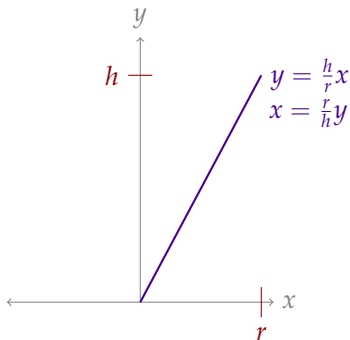
$$V \approx \sum_{i=1}^n (\text{volume of each slice})$$

Consider the volume, V , enclosed by rotating the curve $y = \sqrt{x}$, from $x = 0$ to $x = 4$, around the x -axis.



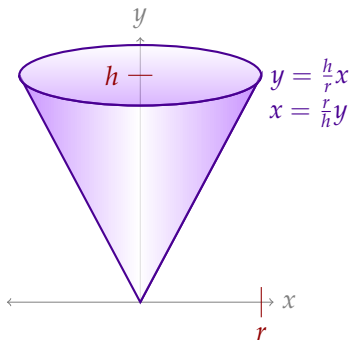
Informally, we think of one slice, at position x , as having thickness dx . So, we can write the volume of this slice as:

Summing up the volumes of slices from $x = 0$ to $x = 4$, our total volume is:



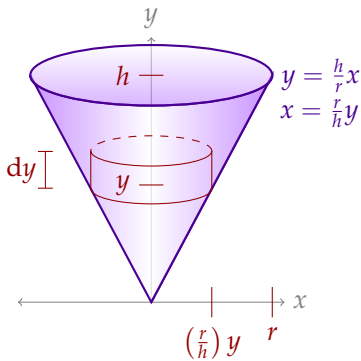
Let h and r be positive constants.

1. What familiar solid results from rotating the line segment from $(0,0)$ to (r, h) around the y -axis?



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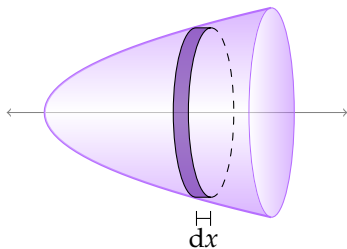
1. What familiar solid results from rotating the line segment from $(0, 0)$ to (r, h) around the y -axis?
2. In the informal manner of the last example, describe the volume of a horizontal slice of the solid taken at height y .
3. What is the volume of the entire solid?

Slice volume: $\pi \left(\frac{r}{h}y\right)^2 dy$

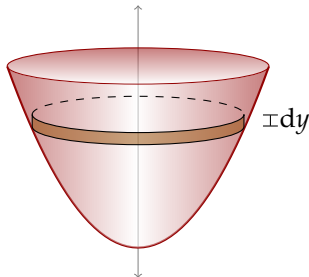
Observation

When we rotated around the **horizontal** axis, the width of our cylindrical slices was dx , and our integrand was written in terms of x .

When we rotated around the **vertical** axis, the width of our cylindrical slices was dy , and we integrated in terms of y .

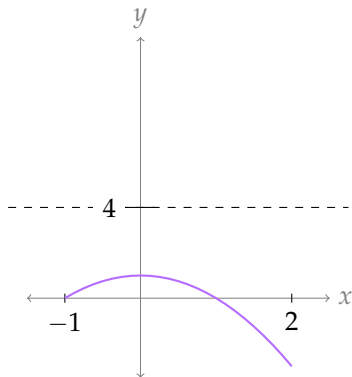


Vertical slices are approximately cylinders



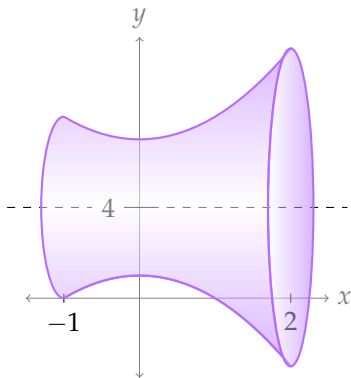
Horizontal slices are approximately cylinders

In this question, we will find the volume enclosed by rotating the curve $y = 1 - x^2$, from $x = -1$ to $x = 2$, about the line $y = 4$.



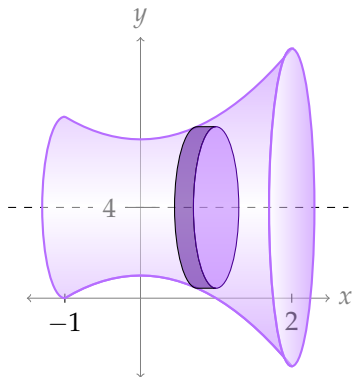
1. Sketch the surface traced out by the rotating curve.
2. Sketch a cylindrical slice. (Consider: will it be horizontal or vertical?)
3. Give the volume of your slice. Use dx or dy for the width, as appropriate.
4. Integrate (with the appropriate limits of integration) to find the volume of the solid.

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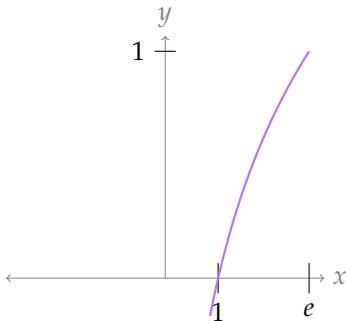


1. Sketch the surface traced out by the rotating curve.
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3. Give the volume of your slice. Use dx or dy for the width, as appropriate.
4. Integrate (with the appropriate limits of integration) to find the volume of the solid.

To find the volume of the entire object, we “add up” the slices from $x = -1$ to $x = 2$ by integrating.

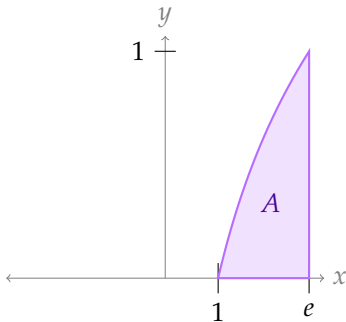
$$\int_{-1}^2 \pi(3 + x^2)^2 dx =$$

Let A be the area between the curve $y = \log x$ and the x -axis, from $(1, 0)$ to $(e, 1)$. In this question, we will consider the volume of the solid formed by rotating A about the y -axis.



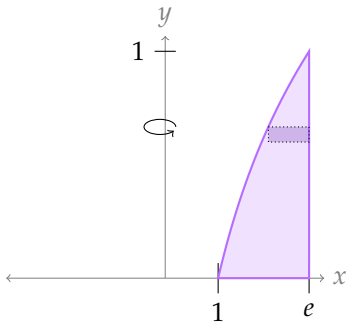
1. Sketch A .
2. Sketch a washer-shaped slice of the solid. (Should it be horizontal or vertical?)
3. Give the volume of your slice. Use dx or dy for the width, as appropriate.
4. Integrate to find the volume of the entire solid.

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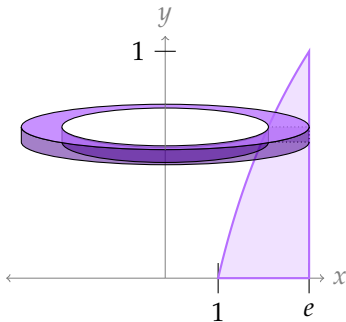
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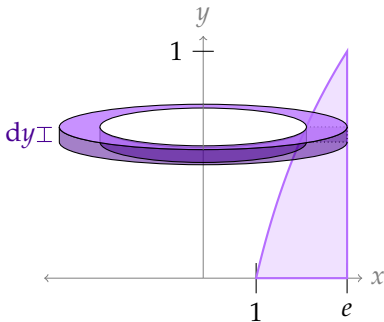
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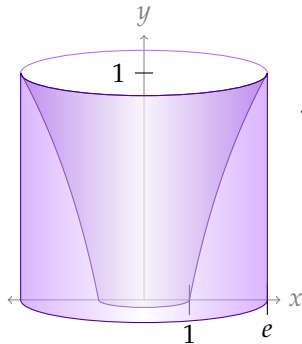
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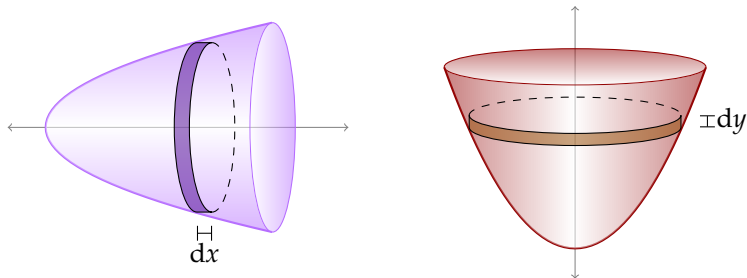
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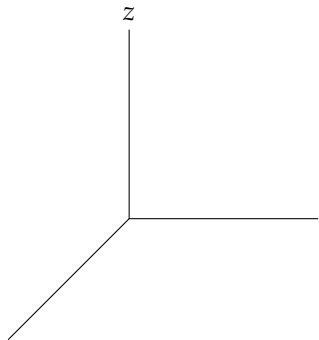
$$\int_0^1 \pi (e^2 - e^{2y}) dy =$$

So far, we've found the volume of solids formed by rotating a curve. When a point rotates about a fixed centre, the result is a circle, so we could slice those solids up into pieces that are approximately cylinders.



We can find the volumes of other shapes, as long as we can find the areas of their cross-sections.

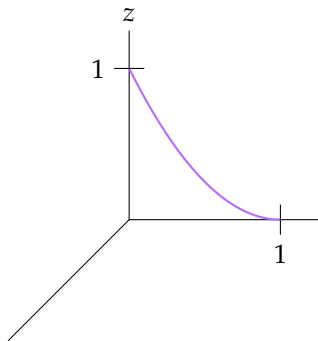
The corner of a room is sealed off as follows:



Q

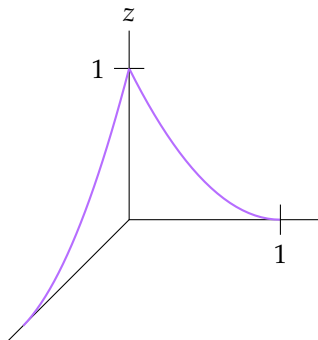
The corner of a room is sealed off as follows:

On both walls, a parabola of the form $z = (x - 1)^2$ is drawn, where z is the vertical axis and x is the horizontal. They start one metre above the corner, and end one metre to the side of the corner.



The corner of a room is sealed off as follows:

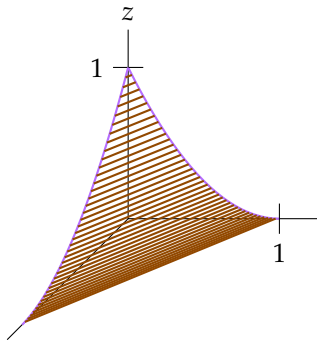
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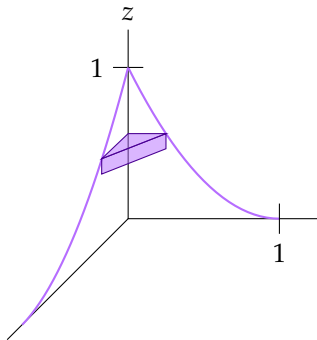
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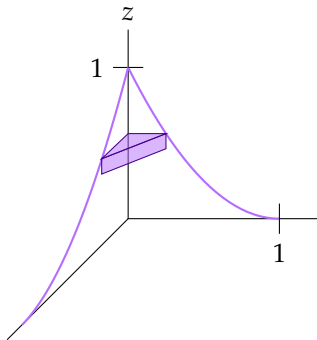
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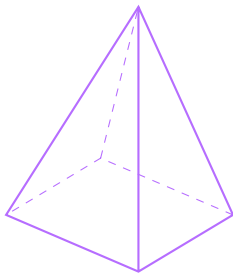
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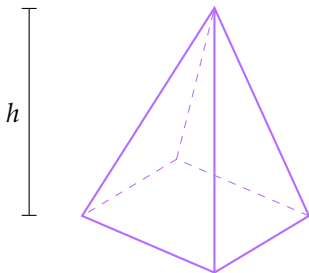
Taught ropes are strung *horizontally* from one parabola to the other, so the horizontal cross-sections are right triangles. **How much volume is enclosed?**



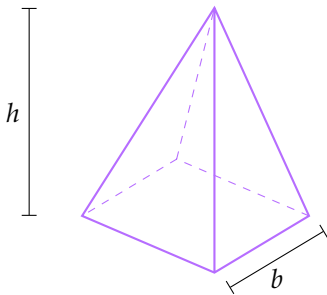
A pyramid with height h metres has a square base with side-length b metres. At an elevation of y metres above the base, $0 \leq y \leq h$, the cross-section of the pyramid is a square with side-length $\frac{b}{h}(h - y)$. What is the volume of the pyramid?



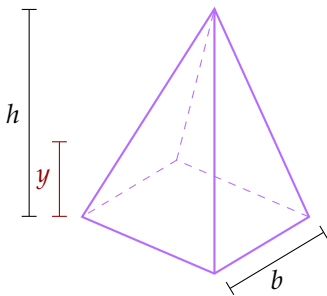
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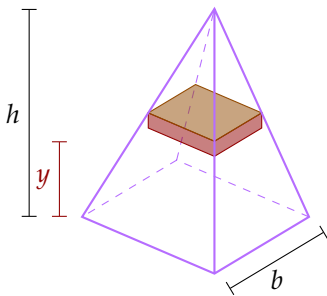
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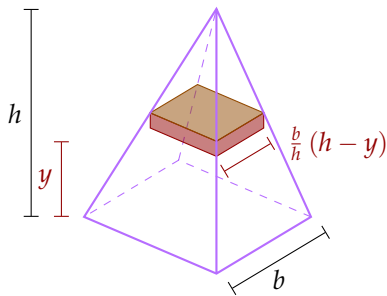
A pyramid with height h metres has a square base with side-length b metres. At an **elevation of y metres** above the base, $0 \leq y \leq h$, the cross-section of the pyramid is a square with side-length $\frac{b}{h}(h - y)$. What is the volume of the pyramid?



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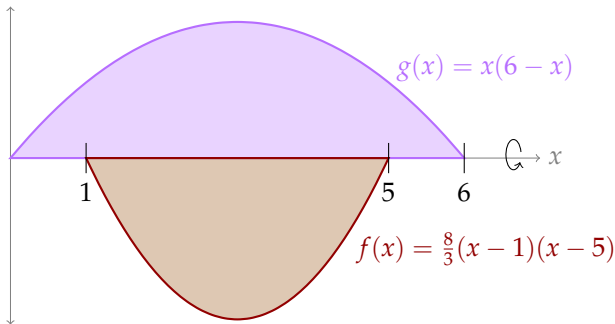
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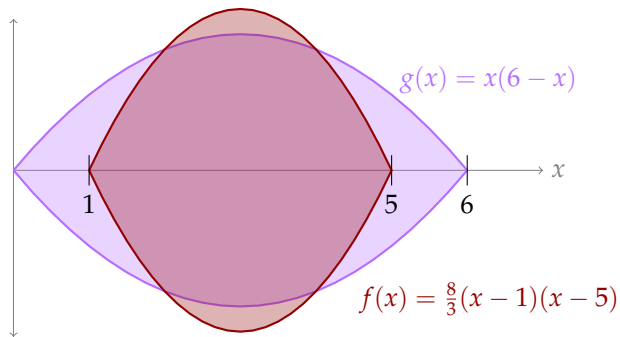
OPTIONAL: CHALLENGE QUESTION

A paddle fixed to the x -axis has two flat blades. One blade is in the shape of $f(x) = \frac{8}{3}(x-1)(x-5)$, from $x = 1$ to $x = 5$. The other blade is in the shape of $g(x) = x(6-x)$, $0 \leq x \leq 6$. The paddle turns through a gelatinous fluid, scraping out a hollow cavity as it turns. What is the volume of this cavity?

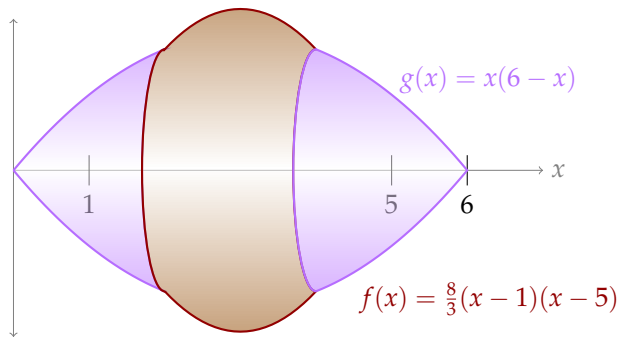
You may leave your answer as an integral, or sum of integrals.



The size of the cavity at a point x along the paddle is determined by the **larger** of $|f(x)|$ and $|g(x)|$.



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The size of the cavity at a point x along the paddle is determined by the **larger** of $|f(x)|$ and $|g(x)|$.

Included Work



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